

Unit 1: Ratios and Proportions

Topic 1: Comprehensive Practice — Classwork 

Name: _____

Date: _____

Q1. What is the ratio of 8 oranges to 12 lemons in simplest form?

- (A) 4:6
- (B) 2:3
- (C) 3:2
- (D) 8:12

Q2. Complete the table: 1:5, 3:15, 6:?

- (A) 20
- (B) 25
- (C) 30
- (D) 35

Q3. A mixture has 5 parts oil and 2 parts gas. How much gas is in 21 liters of mixture?

- (A) 6L
- (B) 15L
- (C) 14L
- (D) 7L

Q4. If the scale of a map is 1cm : 20km, what is the distance for 4.5 cm?

- (A) 80km
- (B) 90km
- (C) 100km
- (D) 45km

Q5. Which pair represents an equivalent ratio to 12:16?

- (A) (6, 9)
- (B) (15, 20)
- (C) (3, 5)
- (D) (18, 22)

Q6. The ratio of $A : B$ is 1:2 and $B : C$ is 3:4. Find $A : C$.

- (A) 1:4
- (B) 3:8
- (C) 2:3
- (D) 3:4

Q7. A rectangle has area 54 sq in and side ratio 3:2. Find the perimeter.

- (A) 30 in
- (B) 15 in
- (C) 20 in
- (D) 60 in

Q8. [FRQ] Write three equivalent ratios for 5:2.

Q9. [FRQ] The ratio of cats to dogs is 5:8. If there are 24 more dogs than cats, how many cats are there?

Q10. [FRQ] x and y are in a ratio 2:3. If $(x+5) : (y+5) = 3 : 4$, find the original values of x and y .

Chef's Tips

- **Mixed Levels:** Start with the easier MQ questions to build confidence before the FRQs.
- **Tape Diagrams:** Use them for the difference-based challenge problems.
- **Logic:** For advanced ratios, always check if your final values simplify back to the original ratio.